

Justin Lee

jl537@rice.edu | (919) 381-2262 | linkedin.com/in/jkl565 | github.com/jbinlee | US Citizen

EDUCATION

Rice University

Houston, TX

Bachelor of Science in Computer Science, Minor in Data Science

May 2027

- **Relevant Coursework:** Object Oriented Programming, Data Structures, Algorithms, Computer Architecture, Multivariable Calculus, Linear Algebra.

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, SQL

Technologies: React, Next.js, FastAPI, Node.js, Pandas, NumPy, Redis, LangChain, Hugging Face

Databases: PostgreSQL, MongoDB, MySQL, LanceDB

EXPERIENCE

OpenStax

Houston, TX

AI Engineer

May 2026 – Present

- Led a **5-engineer** team to architect an interactive branching-scenario application, scaling system design from schema definition to real-time production UI generating **50+** medical training modules.
- Engineered an automated book-ingestion pipeline (**BeautifulSoup**, **markdownify**) converting **5,000+** pages of unstructured textbooks into optimized Markdown, cutting data prep latency from **100+** hours to minutes.
- Designed a model-agnostic backend orchestrating interchangeable LLM endpoints (**Claude/Gemini**) and implemented rigorous prompt tuning to achieve an **80%** reduction in out-of-spec content payloads.
- Architected a **Next.js + FastAPI + PostgreSQL** stack modeling narrative logic as a production directed graph, utilizing custom **React** components to render and mutate AI-generated JSON nodes dynamically.

Scholarhive

Houston, TX

Founding Engineer

October 2024 – January 2025

- Scaled an edtech platform to **5.5K+** users and generated **\$500K** in verified scholarship matches; advanced company to the **YC W25** interview pool[cite: 1].
- Slashed API p95 response latencies by **85%** (from **800ms** down to **120ms**) by designing targeted SQL indexes and deploying high-performance **Redis** caching infrastructure on **AWS**[cite: 1].
- Developed an internal ML-driven matching engine achieving **94%** precision, automating complex multi-tenant data ingest across **10+** distinct sources via an optimized ETL data pipeline[cite: 1].
- Boosted user application conversion rates by **30%** by engineering a highly responsive **Next.js** discovery UI backed by efficient server-side data filtering and custom match-ranking algorithms[cite: 1].

Rice University Digital Health Initiative

Houston, TX

Software Engineer

November 2025 – March 2026

- Engineered a multi-tenant MVP marketplace matching multidisciplinary academic researchers with localized clinical laboratory opportunities using advanced multi-layered eligibility and skill-tagging queries.
- Designed high-throughput data intake channels and fast role-based search pipelines that accelerated cross-institutional lab staffing and dramatically minimized time-to-fill tracking for medical research listings.
- Collaborated directly with Rice faculty and Houston Methodist clinical stakeholders to construct product roadmaps, successfully initiating an early adoption pipeline to secure immediate, active user validation.

PROJECTS

OwlConnect | *LangChain, FastAPI, MongoDB, Next.js, WebSockets, MCP*

September 2025

- Built an agentic matchmaking platform pairing academic cohorts using an embedding model and a **LangChain** multi-agent negotiation system; implemented Next.js, FastAPI, and WebSocket-driven visualizations.
- Awarded **2nd Place Overall** out of **100+** competing engineering teams at HackRice (**500+** total participants).

HootPlanner | *React, TypeScript, FastAPI, Pandas, LanceDB*

September 2024

- Developed an AI-driven course recommendation platform using **Pandas** ETL pipelines, Hugging Face sentence embeddings, and a **LanceDB** vector database backend to achieve low-latency semantic search matching.
- Awarded **1st Place Overall** out of **100+** competing engineering teams at HackRice (**400+** total participants).